

Project Gaea

Value to our planet's nature reserves

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Abstract.

The Earth's ecological balance is under greater threat than ever before. Human economic activity is devastating intact ecosystems at an alarming rate, and this destruction will continue as long as exploitation remains the only recognised economic value of nature. Project Gaea seeks to challenge this paradigm by introducing an asset-backed cryptocurrency rooted in the most fundamental and irreplaceable asset—nature itself. By tokenising nature reserves and creating a tradable currency, we aim to assign nature an economic value that surpasses the value of its exploitation, ensuring that its preservation serves our collective economic interests.

*"You may say I'm a dreamer.
But I'm not the only one."
(John Lennon)*

Introduction

The survival of human civilisation is inextricably linked to the health of our planet's ecosystems. However, untouched natural habitats are vanishing at an alarming rate in the 21st century. In just a few generations, humans have deforested over 50% of Western Europe and Eastern America, leaving only 4.06 billion hectares of forests worldwide, of which a mere 1.11 billion hectares remain as primary forests—the last truly undisturbed ecosystems. Meanwhile, the extinction crisis and accelerating climate change threaten the very foundation of human well-being. The primary driver of this destruction is human economic activity, particularly the expansion of agricultural land and resource extraction. The core issue lies in the fact that intact nature holds little direct economic value for our economic system. Even as we begin to recognise the existential importance of preserving ecosystems, the free-rider problem—where private landowners prioritise short-term gains over long-term environmental benefits—remains a significant barrier to large-scale conservation.

By tokenising nature reserves and creating a tradable currency, we can fundamentally reshape the economic incentives around conservation. Nature reserves can provide two essential financial functions: a store of value and a medium of exchange—just like traditional money. Modern currencies, whether physical (cash) or digital (cryptocurrencies like Bitcoin), derive their utility not from intrinsic worth but from collective trust and acceptance. This principle is now universally accepted in modern societies. However, unlike fiat money, which is issued by governments, or cryptocurrencies like Bitcoin, which exist without tangible backing, a nature-backed currency would be anchored to real, irreplaceable assets: Earth's intact ecosystems.

As of January 2025, the total value of Bitcoin was estimated at \$2 trillion, a seemingly vast sum yet representing just 2.5% of the world's total money supply and 11% of global gold reserves. Even at a conservative valuation, this amount would be enough to purchase 1 billion hectares of nature reserves—

nearly half of the world's remaining rainforest. We hypothesise that tokenising nature reserves can generate greater economic value than their exploitation. By establishing such a currency, we can ensure that the preservation of nature is not only morally imperative but also economically rational.

In the following section, we will introduce GaeaCoin, an asset-backed cryptocurrency (ABC), and explore its core concept. Unlike traditional cryptocurrencies, ABCs require an overseeing institution to manage various aspects of their underlying assets. In the subsequent section, we will outline the structural and operational requirements of Gaea.org, the organisation solely dedicated to maintaining and evolving the ABC.

*"Money makes the world go 'round,
the world go 'round, the world go 'round..."
(Cabaret - Fred Ebb)*

GaeaCoin: A Cryptocurrency with Real Value

A cryptocurrency is a digital asset designed to function as a medium of storage and exchange, where ownership records are securely stored in a distributed ledger. The blockchain is the most widely used form of a distributed ledger, as it solves the fundamental problem of establishing a single, verifiable source of truth in a decentralised system. Cryptocurrencies have proven to be the most successful application of blockchain technology due to their immutability (ensuring past transactions cannot be altered), trustless operation (requiring no trust between users), and decentralised nature (eliminating the need for central institutions). They build on strong cryptographic protocols to secure transaction records, regulate the issuance of new tokens, and verify ownership transfers. The technical challenge of creating trustless digital tokens has already been solved, with the ERC-20 token standard on the Ethereum network serving as the industry benchmark for secure and interoperable tokenisation.

Besides a distributed ledger, successful cryptocurrencies rely on another essential component: a well-designed set of economic incentives that encourage adoption and ensure the system's sustainability. This brings us to the critical question: Why should people use GaeaCoin? Ultimately, GaeaCoin offers three powerful incentives for widespread adoption: (i) a secure, inflation-proof store of value, as its supply is tied to Earth's finite natural habitats rather than arbitrary monetary policies (ii) instant, fee-free transactions through a seamless mobile app, and finally (iii) a moral dimension that exceeds economic interests.

GaeaCoin: Stability Through Asset-Backed Tokenisation

One of the biggest challenges in the crypto market is price volatility, making many digital currencies unsuitable for everyday transactions and unreliable as a store of value. GaeaCoin solves this issue by backing each token with tangible assets - protected nature reserves - mitigating volatility and providing intrinsic stability. The token's price should never drop below the value of its underlying assets. Furthermore, if the price exceeds the asset's value, it incentivises conservation by ensuring the reserves remain untouched. And since its supply is tied to Earth's scarce and finite ecosystems, GaeaCoin should be inflation-proof, unlike fiat currencies which are subject to arbitrary monetary policies.

GaeaCoin: A Truly Fee-Free Digital Currency

Money exchange, whether through digital transactions or cash, comes with hidden costs. Credit card payments incur processing fees of 1–3%, benefiting banks and payment processors while increasing costs for businesses and consumers. International transactions add currency conversion fees and unfavorable exchange rates, while cash handling involves security risks, transport costs, and ATM withdrawal fees. Despite its digital nature, cryptocurrency transactions also incur fees. Bitcoin, for example, operates on a fee market, where users pay miners to process transactions. When network traffic is low, fees can be minimal, but during congestion, they can exceed \$50 per transaction. GaeaCoin eliminates these inefficiencies by offering fee-free transactions, made possible by the non-profit organization Gaea, which operates the necessary blockchain infrastructure without mining fees—a process detailed in the next section.

GaeaCoin: A Currency with a Moral Dimension

Beyond its financial advantages, GaeaCoin carries a powerful ethical dimension: every transaction contributes directly to the protection and preservation of natural habitats, aligning financial gain with environmental responsibility. This innovative approach transforms financial transactions into a direct force for ecological sustainability, allowing users to participate in a global effort to safeguard the planet while engaging in everyday economic activities.

*"If you want to build a ship, don't drum up people to collect wood and don't assign them tasks and work,
but rather teach them to long for the endless immensity of the sea."
(Antoine de Saint-Exupéry)*

Gaea.org: A Decentralized Autonomous Non-Profit Organization

For GaeaCoin to succeed, its supporting organisation, Gaea.org must efficiently provide a wide range of essential services to users. Its mandate must include (i) acquiring and tokenizing nature reserves (ii) monitoring their ecological health and, when necessary, enforcing property rights, (iii) developing and maintaining a suite of software applications, including a blockchain network, smart contracts, and a web/mobile platform (iv) establishing a secure server infrastructure with end-to-end encryption, (v) ensuring legal compliance and managing contractual frameworks, and (vi) executing public relations strategies to foster transparency and adoption. In this section, we will first outline the core principles on which Gaea is founded before detailing how it will deliver the mentioned services.

To ensure that its incentives remain fully aligned with its mission, Gaea.org must function as a non-profit organisation. Furthermore, there are several trust-related aspects to the design of this institution. While cryptocurrencies operate on a trustless system, meaning that no trust is required between users, an asset-backed cryptocurrency demands an equally transparent and decentralised institution to manage its assets and operations. In Project Gaea, we propose a Decentralised Autonomous Non-Profit Organization (DANPO)—an entity that operates without centralised control, using open-source governance, transparent decision-making, and blockchain-based accountability. By ensuring that both the technical and legal framework are fully decentralised, autonomous, and open-source, we empower people worldwide to participate—whether by using the tokens, maintaining the ledger, or contributing to the system's ongoing development.

However, running such an organisation still requires significant financial resources. To remain independent and self-sustaining, Gaea.org will generate revenue through carbon credits. Based on our estimates, the sale of carbon credits derived from tokenised nature reserves will provide a stable, long-term income stream, financing the organisation's operations without relying on external funding or donations.

The carbon credit market is a rapidly growing financial ecosystem designed to incentivise businesses and individuals to reduce greenhouse gas emissions. Carbon credits represent one metric ton of CO₂ removed or avoided, and can be traded on voluntary and compliance markets. Prices vary depending on the type of credit (e.g., afforestation, renewable energy, avoided deforestation), location, and demand, but currently, high-quality voluntary carbon credits trade between \$10 and \$100 per ton, with premium credits (such as those linked to biodiversity conservation) commanding even higher prices. Given that one hectare of well-preserved rainforest can sequester between 10 and 30 metric tons of CO₂ per year, a 100-hectare reserve could generate between \$10,000 and \$300,000 annually from carbon credit sales alone. As demand for nature-based solutions continues to rise—driven by corporate sustainability commitments, net-zero pledges, and regulatory pressures—prices are expected to increase further, making carbon credit revenues the key financial pillar for project Gaea. Any income that exceeds the organisation's costs is used to acquire new property and distributed as dividends to the token holders. By tokenising natural assets and selling the carbon credits, the project can create a self-sustaining economic model, where the preservation of ecosystems becomes a profitable and scalable investment.

Furthermore, by sequestering carbon from the atmosphere nature will essentially provide a currency and payment system free of charge to all humans. By simply using this currency to save and exchange money, humans will protect and safeguard earth's remaining nature - essentially balancing economic growth and intact ecosystems.

*"Unless someone like you
cares a whole awful lot,
nothing is going to get better.
It's not."
(The Lorax - by Dr. Seuss)*

Acquisition

Gaea will provide a website, on which land owners, real estate agents, and the general public can interact to fundraise the acquisition of new nature reserves in a crowdfunding scheme. This will enable individuals from around the world to collectively purchase and protect natural habitats in a transparent and trustless manner using blockchain technology. Through the platform, backers can contribute funds to specific conservation projects, and once a funding goal is met, the system will automatically execute smart contracts to purchase and tokenise the land. These smart contracts ensure full transparency, as every transaction is recorded on the blockchain, allowing contributors to track exactly how their funds are allocated. Each purchased property is minted into tokenised shares, representing fractional ownership or conservation rights, which are then distributed to backers' wallets. Furthermore, token holders can participate in governance decisions, such as voting on reinvesting revenue from carbon credit sales into new acquisitions. This innovative model democratises conservation, allowing anyone to play a direct role in protecting ecosystems

while benefiting from an economically sustainable system that rewards long-term environmental stewardship.

Tokenomics

GaeaCoin operates on a multi-layered system of abstraction, ensuring a transparent and verifiable connection between natural reserves and digital assets. This structure allows for an effective representation of ecological value within the blockchain ecosystem.

1. The Foundation: Natural Reserves

At the core of GaeaCoin's tokenomics are natural reserves, which include various forms of protected land such as primary forests, wetlands, and other ecologically significant areas. These reserves serve as the fundamental basis for the tokenisation process.

2. Property Acquisition: Purchase Agreement

The process of tokenising a novel asset begins with writing all relevant information concerning this asset into a Proof-of-Asset (PoA) card, which will be written into a decentralised public database. This ensures complete transparency in the process and allows the general public and the token holders in particular to completely oversee all assets underlying the tokens. The PoA card will contain the following data: (i) a time stamp of creation, (ii) the purchase agreement, (iii) the extract from the land register providing information about the land area, the owner, notes, charges, etc., (iv) the GPS coordinates, (v) a report by an ecologist certifying the value of the land with respect to preservation, (vi) a report on the enforceability of its preservation, (vii) satellite images documenting the initial state of the reserve, and (viii) the contact of the legal representation and security service for the enforcement of property rights. Through a purchase agreement, the land registry transfer is completed, officially placing the property under the ownership of a designated NGO. This ensures that the land is protected and managed in alignment with conservation goals.

3. Tokenisation: ERC-721 Certificate

At the second level, an ERC-721 non-fungible token (NFT) certificate is issued by the NGO. This certificate represents the property on the blockchain and determines the number of tokens that can be minted from it. Gaea will provide a formula according to which the exchange rate between the proposed nature reserve and the GaeaCoin is established. This formula will include (i) the size of the property, (ii) its geographical location, (iii) its carbon sequestration potential, (iv) its ecological value, including the state of flora and fauna on the property, and (v) the presence of endangered species. The ERC-721 certificate ensures that each property's ecological significance is properly quantified and transparently recorded.

4. Fungible Tokens: ERC-20 Representation

At the highest level of abstraction, the system generates ERC-20 fungible tokens, which are minted via a Minter Smart Contract (MSC). These tokens are divisible and not uniquely tied to any certain asset. Instead, their total supply is derived from the corresponding ERC-721 certificate's valuation. This fungible structure allows for fluid trading, liquidity, and broad adoption while maintaining a direct link to real-world conservation efforts.

Monitoring and Preserving

The continuous monitoring of the integrity of the nature reserve will be based on satellite images. We will use a machine learning model to detect possible anomalies and feed both the images as well as the model output into the blockchain with a suitable cryptographic hash function. This renders the process completely transparent to all stakeholders. Copernicus, the European Union's Earth observation programme ([www.copernicus.eu](<https://www.copernicus.eu/en>)), could serve as the required blockchain oracle, a third-party service that connects smart contracts with the outside world, primarily to feed external information. It provides trusted APIs for accessing images based on GPS coordinates. Daily updated images will be evaluated by an anomaly detection model e.g. based on a convolutional neural network. The computational work involved in applying the model to the satellite images should be part of the mining process substituting otherwise meaningless cryptographic work. A detected anomaly in one of the images will prompt human inspection, implemented by a smart contract. The verification of the anomaly by an expert will trigger security and law enforcement. The preservation of nature reserves involves physical presence in the designated area. This includes enforcing property rights by means of legal and security services. Thus, every asset will have an assigned legal representation as well as a security service enforcing its property rights in the worst case.

Legal services and public relations.

Running an asset-backed cryptocurrency like GaeaCoin requires a strong foundation of legal services and public relations to ensure compliance, credibility, and widespread adoption. From a legal perspective, the organisation must navigate property law, financial regulations, and environmental protections across multiple jurisdictions, ensuring that land acquisitions, tokenisation, and carbon credit sales are legally enforceable. This includes drafting and managing contracts for land purchases, securing conservation easements, and structuring legally binding agreements that define token holders' rights. Additionally, the organisation must establish regulatory compliance mechanisms, particularly in relation to securities law, anti-money laundering (AML) regulations, and tax obligations, ensuring that the cryptocurrency operates within legal boundaries while maintaining decentralisation. A dedicated legal team will also be crucial for resolving disputes, protecting intellectual property, and ensuring that smart contracts align with real-world legal frameworks.

On the public relations (PR) side, the organisation must engage in strategic communication campaigns to build trust among investors, environmental organisations, and policymakers. This involves media outreach, content creation, and partnerships with sustainability advocates, positioning the cryptocurrency as a legitimate financial instrument for conservation. Transparent reporting, regular updates on project milestones, and community engagement will be key to maintaining credibility. Additionally, crisis management strategies must be in place to address regulatory challenges, misinformation, or market volatility, ensuring that the organisation can respond swiftly and effectively. By integrating robust legal frameworks with proactive public relations, GaeaCoin can establish itself as a trustworthy, legally sound, and widely accepted solution for sustainable asset-backed finance.

*"The greatest threat to our planet is the belief that someone else will save it." *